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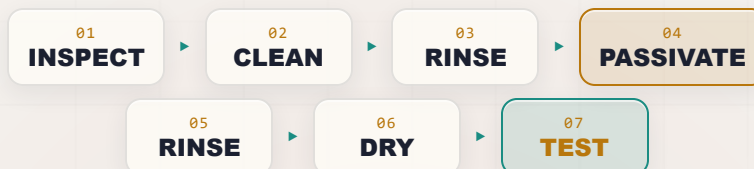
PASSIVATION (STAINLESS S

THE COMPLETE TRAINING MANUAL

SURFACE ENHANCEMENT • REMOVES FREE IRON • NO COATING ADDED

A full training course for the brand-new operator — from “wait, if nothing gets added, what is passivation?” to a signed certificate of completion. The chemical treatment that cleans free iron off stainless steel and lets the steel’s own chromium-oxide passive layer reform stronger and more uniform. No plating tanks, no deposited film, no mystery. Assumes you know nothing.

Teaches you everything.



OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 • 2026 • FOR-DUMMIES STYLE

Training reference only. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer and industry specifications (ASTM A967, AMS 2700), current Safety Data Sheets (SDS), and applicable OSHA, EPA, REACH, and local regulations before production use.

— FIND YOUR WAY

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READ FRONT-TO-BACK THE FIRST TIME · KEEP ON THE SHELF FOREVER AFTER

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REMEMBER

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, the governing customer or industry spec, or your supervisor's instructions. When this manual and your shop's official procedure disagree, **follow your shop's official procedure and ask your supervisor.**

WELCOME ABOARD

IF YOU'VE NEVER HEARD THE WORD “PASSIVATION” BEFORE, YOU ARE IN EXACTLY THE RIGHT PLACE

Welcome. If you're holding this manual, you're about to learn how to run — or at least understand — a **passivation line**: the chemical treatment that cleans and strengthens the surface of **stainless steel** parts so they resist rust and corrosion the way stainless is *supposed* to. Maybe you started this week. Maybe you've been loading baskets for a year and finally want to know *why* you do the things you do. Either way, you're in the right place, and we're glad you're here.

Here's the most important thing we can tell you up front: **this manual assumes you know nothing about chemistry, metallurgy, or why stainless steel doesn't rust**. That's not an insult — it's a promise. Every term gets defined the first time we use it. Every step explains not just *what* to do but *why* it matters. Nobody is born knowing what “free iron” or “the passive layer” means. You learn it. We're going to teach it — in the spirit of the *For Dummies* books: friendly, plain-spoken, and patient.



REMEMBER

You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday.



HEADS-UP

One big idea before we start, and it's the single most important reframe in this whole book: **passivation does not add a coating to your parts**. There is no film deposited, no metal plated, no layer painted on. Passivation *cleans* contamination (especially “free iron”) off the stainless and then lets the stainless steel's *own* natural protective skin reform stronger and more uniform. And do not let “no coating” fool you into thinking the line is harmless: you'll work with **nitric acid** (a strong oxidizer that can throw off toxic brown fumes), hot acid baths, alkaline or solvent degreasers, and hot rinses. Respect all of it.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: inspect it, clean it, rinse it, passivate it, rinse it, dry it, and test it. Reading it in order matters, because (as you'll soon learn) **the order of a passivation line is not random** — and neither is the order of this book.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

As you read, you'll run into little flagged notes. Here's what each one means.



TIP

A shortcut or trick of the trade — the kind of thing a 20-year veteran would lean over and tell you.



REMEMBER

A core idea worth locking in. If you forget everything else on the page, don't forget these — they're the load-bearing ideas.



HEADS-UP

A safety warning. Read every one — these keep your skin, eyes, and lungs intact. We use them *liberally*, because nitric acid is no joke.



TECHNICAL STUFF

Deeper chemistry for the curious. Totally optional — skip it and you can still run the line perfectly.



FUN FACT

A bit of trivia to make the science stick (and give you something to say at lunch).

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS FOUNDATION AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what passivation is** in plain language — a chemical *cleaning and enhancement* process — and why it is **not** a coating, plating, or conversion coating.
2. **Describe the two things passivation accomplishes:** it removes **free iron** and surface contamination, and it lets the steel's own **chromium-rich oxide “passive layer”** reform, thicken, and become more uniform.
3. **Explain where corrosion resistance actually comes from** — the chromium in the stainless, not from anything we add — and why free iron ruins it.
4. **Name the two main chemistries** — **nitric acid** (classic) and **citric acid** (modern, greener, safer) — and the tradeoffs of each.
5. **Name all seven stations of the line in order** and explain *why* the order cannot be rearranged — especially why **degreasing comes first** and **testing comes last**.
6. **Recognize the main hazards** — nitric acid burns and **NOx (brown) fumes**, hexavalent chromium if dichromate is used, hot acid baths, degreasers, slips — and the correct PPE and response for each.
7. **Understand grade-dependence and flash attack** — why free-machining and high-carbon/martensitic stainless need gentler conditions than forgiving austenitic 300-series.
8. **Verify the work by testing** — because there's no visible coating, you confirm success with water immersion, humidity, salt spray, copper sulfate, and ferroxyl tests.
9. **Tell passivation apart from pickling** — and know that heavy scale or weld discoloration is a *pickling* job, not a passivation job.



REMEMBER

You are not expected to hit all of these on day one. Surface finishing is a craft. The goal is steady competence, not instant mastery.

• THE SEVEN-STATION PROCESS AT A GLANCE

Here's the whole passivation line in one strip. Don't memorize it yet — just get a feel for the rhythm: **Inspect** → **Clean** → **Rinse** → **Passivate** → **Rinse** → **Dry** → **Test**.



REMEMBER

Notice the bookends: the line *starts* with cleaning (because passivation fails on a dirty part) and *ends* with testing (because passivation leaves nothing visible — the test is how you know it worked). A rinse follows the clean and follows the passivate, keeping one chemistry from poisoning the next.

CHAPTER 1

WHAT IS PASSIVATION?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHY STAINLESS RESISTS RUST

THE ONE-SENTENCE VERSION

Passivation is a chemical treatment — usually a nitric-acid or citric-acid immersion — that removes free iron and other contamination from a stainless steel surface and lets the steel's own chromium-rich oxide film reform stronger, thicker, and more uniform. That's the whole idea. Now let's unpack it, because the most common misunderstanding in the whole field is hiding in that sentence: **nothing is added to the part.**

THE BIG REFRAME: NO COATING IS ADDED

If you've seen our other manuals, you've met processes that *add* something to a part. Passivation is the odd one out — and understanding the difference is the key to everything.

- **Electroplating** *adds* a layer of a *different* metal on top of the part, using **electricity** (for example, zinc or nickel onto steel). The result is a brand-new metal skin sitting *on* the original surface.
- **Conversion coatings** (like iron phosphate or chromate) *convert* the top of the surface *into* a new chemical compound — a different material than the bare metal — that then becomes the coating.
- **Passivation** (this manual) *adds nothing and creates no new coating material*. It **cleans** the stainless surface (dissolving away free iron and contamination) and then lets the stainless steel's **own, naturally occurring** chromium-oxide film reform and strengthen. The corrosion resistance was always *inside* the stainless — passivation just removes what was blocking it.



REMEMBER

Lock in the difference: **Plating adds a new metal. A conversion coating turns the surface into a new compound. Passivation adds nothing — it cleans off the iron and lets the steel protect itself.** If you ever go looking for the “film we deposited” after passivation, you won't find one. That's not a missing piece — that's the entire point. The protection comes from the stainless steel's own chromium, not from us.



FUN FACT

The passive layer is so thin that if a sheet of stainless were as tall as a 30-story building, the protective film on it would be about the thickness of a sheet of paper. You can't see it, you can't feel it, and yet it's the only thing standing between your stainless and a rusty mess.

• WHY STAINLESS STEEL RESISTS RUST IN THE FIRST PLACE

Ordinary carbon steel rusts because there's nothing protecting the iron. Stainless steel is different because it contains **chromium** — at least about **10.5%** by weight. That chromium reacts with oxygen in the air (or water) to form an incredibly thin, invisible, tightly-bonded film of **chromium oxide** right on the surface. That film is called the **passive layer** (or passive film). It's only a few atoms thick, it's transparent, and it has a magic property: if you scratch it, it *heals itself* by grabbing more oxygen — as long as the surface is clean and has enough exposed chromium. That self-healing chromium-oxide skin is what makes “stainless” stainless.

SO WHAT DOES THE ACID ACTUALLY DO?

Two things at once — and this is the heart of the chemistry:

1. **It dissolves free iron and contamination.** The acid preferentially attacks and dissolves iron and other contaminants sitting on the surface, washing them away.
2. **It enriches and rebuilds the chromium oxide.** As the iron is stripped away, the surface is left **richer in chromium** relative to iron. Nitric acid (and the oxidizers present) then helps **grow and thicken** the chromium-oxide passive film — faster and more uniformly than air alone.

HOW PASSIVATION WORKS — FREE IRON REMOVED, NATIVE OXIDE REFORMS (NO FILM DEPOSITED)



Nothing is added — the steel's own chromium-oxide layer reforms cleaner and stronger.



HEADS-UP

“No coating added” does **not** mean “no hazard.” The classic chemistry is **nitric acid** — a strong oxidizer and corrosive that causes severe burns and, when it attacks metal too aggressively, throws off **brown/red-brown NOx fumes** that are dangerous to breathe. The whole line still needs ventilation, PPE, and respect. We'll hammer this throughout.

THE PASSIVE LAYER, FREE IRON & THE TWO CHEMISTRIES

WHY A TINY BIT OF STRAY IRON CAN RUIN GOOD STAINLESS — AND THE TWO ACIDS WE USE TO FIX IT

• FREE IRON: THE VILLAIN OF THIS STORY

If stainless protects itself with a chromium-oxide film, why does it ever rust? Usually the answer is **free iron** (also called *exogenous iron* or surface iron contamination): tiny bits of ordinary, non-stainless iron left *on* the stainless surface from manufacturing. Where does it come from?

- **Machining, grinding, and cutting** with tools that also cut carbon steel — particles of carbon-steel iron get smeared and embedded into the softer stainless surface.
- **Forming, stamping, and bending** on dies that have handled carbon steel.
- **Steel wool, carbon-steel brushes, or sanding media** used on the stainless (a classic mistake).
- **Contact with carbon steel** — laying stainless parts on a steel bench, hauling them in steel bins, even handling with bare hands over time.

That free iron is *not* stainless — it has no chromium to protect it. So it **rusts**. And worse, a rust spot can become a launch pad for **localized corrosion** (pitting) that eats into the good stainless underneath. A few specks of free iron can make a perfectly good stainless part fail in service.



REMEMBER

Free iron is the enemy. It's ordinary iron stuck onto the stainless during manufacturing. It rusts, it looks terrible, and it can trigger pitting in the good steel around it. Passivation's #1 cleaning job is to **dissolve that free iron away** so only chromium-rich stainless is left at the surface — free to form its protective film.



FUN FACT

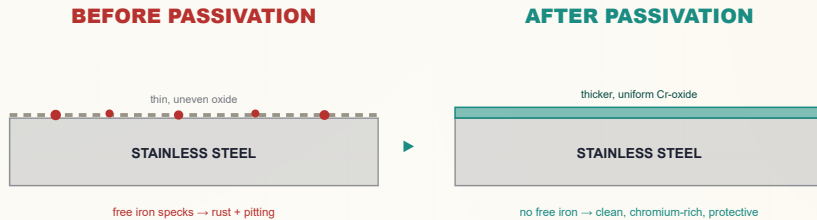
A brand-new stainless part can fail a salt-spray test and grow rust spots within days — not because the stainless is bad, but because invisible carbon-steel particles from the machine shop are sitting on it. Passivate the same part and those rust spots vanish — not because we coated over them, but because we *dissolved the contamination that was rusting*.

• THE PASSIVE LAYER: THE HERO

The **passive layer** is the thin, self-healing chromium-oxide film that stainless forms on its own. Passivation makes that film the best it can be:

- **Cleaner** — no free iron, oil, or debris interfering with it.
- **More chromium-rich** — because the acid removed surface iron, leaving relatively more chromium.
- **Thicker and more uniform** — because the acid/oxidizer accelerated and evened out the film's growth instead of waiting on slow air exposure.

SAME STEEL, BETTER SURFACE — NOTHING ADDED



Same part, same alloy — we removed the iron and let the steel's own film come back better.



TIP

Stainless will *eventually* form a decent passive film just sitting in clean air — but “eventually” can take days or weeks, it’s uneven, and it does nothing about embedded free iron. Passivation does in 20–30 minutes, reliably and uniformly, what nature does slowly and imperfectly — *and* it removes the free iron that air exposure never could.

• THE TWO CHEMISTRIES

There are two main passivation chemistries. You'll run one or both. Know which your shop uses.

1. NITRIC ACID (THE CLASSIC)

A bath of **nitric acid** in water — typically in the **~20–50% by volume** range — used for decades and still the reference for many specs. For certain hard-to-passivate grades, a small amount of **sodium dichromate** is added to suppress over-aggressive attack ("flash attack").

- **Strengths:** proven, well-understood, strong oxidizer that builds a robust passive film, effective across many grades (with the right type).
- **Weaknesses:** **hazardous** — strong oxidizer, severe burns, and it can evolve toxic **NOx (brown) fumes**, especially if it attacks the steel. **Sodium dichromate is hexavalent chromium (Cr(VI)) — a confirmed carcinogen.** Nitrate and chromium in the waste stream are regulated.

2. CITRIC ACID (THE MODERN, GREENER CHOICE)

A bath of **citric acid** in water — typically **~4–10% by weight**, often warm. Citric acid is the same mild acid found in lemons; in industry it's a safer, environmentally friendly chemistry that's **increasingly the preferred standard**.

- **Strengths:** **much safer and greener** — biodegradable, far less hazardous, **no NOx fumes**, no Cr(VI). Gentler on sensitive grades (less flash-attack risk). Easier waste handling.
- **Weaknesses:** baths can grow biological/mold contamination if not maintained; some legacy customer specs still *call out nitric specifically*, so you may not always be free to substitute.



REMEMBER

Nitric is the classic but hazardous one (NOx, and Cr(VI) if dichromate is added). Citric is the modern, safer, greener one (biodegradable, no NOx). The industry is steadily moving toward citric for safety and environmental reasons — but always follow the chemistry your **customer's spec** allows. Some specs let you choose; some demand a specific method.



REMEMBER

Passivation is not a coating, and it is not a miracle. It can't fix weld scale, heat tint, or a bad alloy. Heavy scale must be **pickled or mechanically removed** *before* passivation — passivation is the gentle finish, not the heavy lifting. (Whole section on pickling vs. passivation in Part Three.)

CHAPTER 3

HOW A PASSIVATION LINE WORKS

THE BIG PICTURE — A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A LINE IS A SEQUENCE OF STAGES

A passivation line is not one tank. It's a **sequence of stages** that a part travels through in a strict order, like a dishwasher cycle: each station does one job, and each job prepares the surface for the next. Parts move through in two common ways:

- **Immersion (dip), most common for passivation.** Parts (in baskets, racks, or barrels) are lowered into a series of tanks in turn — degrease, rinse, passivate, rinse, dry — then tested. Gentle, even, and great for complex shapes, threads, and blind holes where the passivating acid needs to flood every surface.
- **Spray/cabinet or manual application** for very large or fixed items — the chemistry is sprayed, flooded, or wiped on. Less common; same chemistry, different delivery.



TIP

Passivation contact time is *long* compared with many finishing steps — often **20–30 minutes or more** in the passivating bath. That's normal and necessary: dissolving embedded free iron and growing a uniform passive film takes time. Resist the urge to rush it; "just a quick dip" does not passivate.

• THE WHOLE SEQUENCE: 7 STATIONS (TEACHING LAYOUT)

Here's the complete process as we'll teach it. Don't memorize the details — just get a feel for the *flow*. Each station gets its own full chapter.

#	STATION	THE ONE THING IT DOES	TEMP	TIME
01	Inspect & Load	Confirms the part & alloy; racks it so acid reaches everything	Ambient	–
02	Clean / Degrease	Removes oil/soil so the acid can reach bare stainless (CRITICAL)	Amb-180°F	5-15 min
03	Rinse (Post-Clean)	Washes off degreaser before the acid stage	Ambient	1-2 min
04	Passivate (Nitric/Citric)	Dissolves free iron & enhances the chromium-oxide passive layer	70-160°F	20-30+ min
05	Rinse (Post-Passivate)	Washes off all acid; stops the reaction; protects the surface	Ambient	1-2 min
06	Dry	Removes all water to prevent water spotting/staining	Amb-250°F	As needed
07	Test & Verify	Confirms it worked — because there's no coating to see	Per test	Per test

The basic rhythm: **Inspect → Clean → Rinse → Passivate → Rinse → Dry → Test.**

**REMEMBER**

Two stations carry special weight on a passivation line. **Station 02 (Clean)** because passivation simply *fails* on an oily surface — the acid can't reach the steel it needs to clean. And **Station 07 (Test)** because passivation leaves *nothing you can see* — unlike a coating you can eyeball, the only way to know passivation worked is to **test for it**. Cleaning at the front, testing at the back: those are the bookends that make the whole thing trustworthy.

• A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

**REMEMBER**

Every stage either protects the next stage or sabotages it. There is no neutral step. A part that leaves the degreaser still oily will not passivate. A part that leaves the acid bath un-rinsed will carry acid forward and stain or flash-rust. A part that's tested wet or dirty gives a false result. The line is a chain, and a chain is only as strong as its weakest link.

**TIP**

Think of the passivate stage as the heart of the line. Everything *before* it (inspect, clean, rinse) exists to deliver a perfectly clean, bare, well-rinsed stainless surface for the acid to work on. Everything *after* it (rinse, dry, test) exists to protect and *prove* what the passivation accomplished. The whole line orbits that one stage.

**HEADS-UP**

This seven-station layout is the *teaching* version. Your shop may combine the dry and test, run an extra rinse, add a separate alkaline-derust or citric-with-passivator step, or use an immersion line versus a basket/barrel setup. Both immersion and spray/soak approaches are covered. Always run *your* line's actual sequence and posted parameters.

**FUN FACT**

Because passivation deposits nothing, a perfectly passivated part and a completely untreated part can look *identical* sitting side by side on the bench. That's exactly why the line ends in a test station instead of a visual inspection — the proof of a passivation job is what the surface *does*, not how it looks.

CHAPTER 4

WHY THE ORDER MATTERS

WALKING THE LINE — EACH STAGE SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY

You might think the stage order is just convention — that you could shuffle steps or skip one to save time. You can't. Each stage exists *because of* the stage before and after it. Here's the logic, link by link.

STATION 02 — CLEAN: GET THE OIL OFF FIRST (OR NOTHING ELSE WORKS)

The passivating acid must touch *bare stainless*. Oil, grease, coolant, fingerprints, and shop soil block it completely. **Passivate over oil and you've passivated nothing** — the acid can't reach the free iron it's supposed to dissolve, and the result is a part that *looks* done but still rusts. There's a free, foolproof check called the **water-break test**: pull a cleaned part and watch the water. If it sheets off in an unbroken film, the surface is clean. If it beads up like rain on a waxed car, oil remains — re-clean.



REMEMBER

Degreasing is not a warm-up — it is the make-or-break step. More failed passivation jobs trace back to inadequate cleaning than to anything in the acid bath. A passivation line lives or dies at Station 02. If you remember one thing about order: *clean comes first, and clean must be complete.*



HEADS-UP

Depending on your shop, the cleaner may be a **hot alkaline degreaser** (caustic — chemical burns, eye injury) or a **solvent** (flammable and/or an inhalation hazard). Know which you run, wear the right PPE, and know where your eyewash and shower are *before* you need them.

STATION 03 — RINSE (POST-CLEAN): DON'T CARRY THE DEGREASER FORWARD

If the cleaner is **alkaline** (high pH) and the passivating bath is **acidic** (low pH), they're chemical opposites — drag alkaline cleaner into the acid and you neutralize and weaken it, pushing it out of spec. Even a solvent or neutral cleaner leaves residue that interferes. The rinse strips the cleaner film off first.



REMEMBER

"Drag cleaner into the acid and you blunt the acid." Rinses exist to keep each chemistry from invading the next stage. This is your first firewall on the line.

STATION 04 — PASSIVATE: THE MAIN EVENT

Now — and only now, with a clean, bare, well-rinsed stainless surface — the acid does its work: it dissolves free iron and contamination and helps the chromium-oxide passive layer reform stronger and more uniform (Chapters 1–2 covered the chemistry). This is the payoff stage.



HEADS-UP — NITRIC NOX

If this is a **nitric** bath, it is a **strong oxidizer and corrosive**. Watch for **brown/red-brown fumes (NOx — nitrogen oxides)** rising off the bath, especially if the acid is attacking the steel (flash attack) instead of just cleaning it. **Those fumes are a serious inhalation hazard** — they can cause delayed, life-threatening lung injury (pulmonary edema) hours later. **Local exhaust ventilation is required**, and nitric must be kept away from organics, solvents, and rags (violent reactions). If you see brown fumes, that's a red flag — ensure ventilation, do not breathe it, and report it. (Citric baths produce no NOx — one of the big reasons shops switch.)

STATION 05 — RINSE (POST-PASSIVATE): STOP THE REACTION, PROTECT THE SURFACE

The part leaves the acid bath wet with spent acid and dissolved iron. **Rinse it thoroughly** to stop the reaction, wash away the dissolved contamination, and prevent acid from drying on the surface (which causes staining and can even *re-attack* the steel). The last rinse before drying ideally uses **clean or deionized (DI) water** so no mineral residue spots the surface.



REMEMBER

Acid left to dry on stainless stains it — and can undo your work. This rinse is the wall between a clean, passivated surface and a stained, re-contaminated one. Skimp here and parts can streak, spot, or flash-rust.

STATION 06 — DRY: NO WATER SPOTS, NO STAINING

Water left to evaporate on its own leaves mineral spots and can even start surface staining. The dry station removes all water — by warm air, an oven, or clean wipe-down — leaving a clean, dry, spot-free surface.



REMEMBER

A passivated part that air-dries with hard-water droplets can come out covered in spots that look like a defect. Proper drying (ideally after a DI final rinse) is what makes the finish *look* as good as it *is*.

STATION 07 — TEST & VERIFY: PROVE IT (BECAUSE YOU CAN'T SEE IT)

This is the station that makes passivation different from every coating process. There is **no film to inspect, no color to read**. The only way to know the part is properly passivated is to **test it** — with a water immersion, high-humidity, salt-spray, copper sulfate, or ferroxyl test (covered in depth in Part Three). The test is your proof.



REMEMBER

With a coating, you can see the result. With passivation, you test for it. The verification test isn't paperwork — it's the *only* way anyone, including the customer, knows the job worked. Treat the test station as seriously as the acid bath.



REMEMBER

The whole seven-station sequence is one continuous logic chain: *clean it so the acid can reach bare stainless → rinse so the cleaner doesn't blunt the acid → passivate to remove free iron and rebuild the passive layer → rinse so acid doesn't dry and stain → dry so no water spots → test so you know it worked*. Every step earns its place. Skip one, and the part — or the customer — tells on you.



TIP

When you understand *why* each stage exists, troubleshooting becomes logical instead of mysterious. Part still rusts or fails the test? Look upstream at cleaning and the acid bath. Stains and streaks? Look at the post-passivate rinse and the dry. Weld area still corroding? That's not a passivation problem at all — it's heat tint that needed pickling first. The defect almost always points back to the stage that was supposed to prevent it.



FUN FACT

The order "clean, then acid, then test" isn't unique to passivation — it's the same logic a hospital uses: prep the field, do the procedure, then verify the result. On a passivation line, the "verify" step is non-negotiable precisely because the work is invisible.

CHAPTER 5

PPE — YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND ACIDS THAT BURN, FUMES THAT INJURE LUNGS, HOT BATHS THAT SCALD

A passivation line works with nitric or citric acid, possibly hot acid baths, alkaline or solvent degreasers, hexavalent chromium (if dichromate is used), acid mist and **NOx fumes**, and slick wet floors. None of these will hurt you if you respect them and wear the right gear. Several can hurt you badly if you don't.

★

REMEMBER
There is no part, no order, and no deadline worth an injury. Scrap can be reworked. Your eyes and lungs cannot.

• THE MASTER PPE SET (WEAR ON THE LINE)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; a splash finds the gaps.
- **Face shield** — over the goggles for any work involving the acid bath, the degreaser, or charging/dumping tanks.
- **Chemical-resistant gloves** — elbow-length, rated for **nitric acid** (a strong oxidizer — verify glove compatibility; nitrile may degrade in concentrated nitric) and for the degreaser. Inspect for pinholes before each use.
- **Chemical-resistant apron** — full-length, chest to below the knee.
- **Chemical-resistant, non-slip boots** — floors around acid and rinse tanks are wet and slick.
- **Long sleeves / appropriate clothing** — no exposed skin where splashes or mist are likely.
- **Respiratory protection** — required when ventilation is inadequate or when working over a fuming nitric bath. Use the cartridge or supplied-air respirator your shop specifies.

⚠

HEADS-UP — NITRIC & NOX
Nitric acid mist and **NOx (brown) fumes** are a primary inhalation hazard on this line. **Local exhaust ventilation is your first line of defense; respirators back it up.** If you can see brown fumes or smell a sharp acid odor, ventilation isn't keeping up — stop and report it. NOx exposure can feel mild at the time and cause serious lung injury *hours later* — never tough it out.

⚠

HEADS-UP — HEXAVALENT CHROMIUM
If your nitric bath contains **sodium dichromate**, you are handling **hexavalent chromium (Cr(VI))** — a confirmed human carcinogen regulated by OSHA under **29 CFR 1910.1026** (low exposure limit, monitoring, respiratory protection, medical surveillance, strict housekeeping). This goes far beyond ordinary line PPE. Know whether your bath contains dichromate. Citric and plain-nitric baths do not.

• STATION-BY-STATION HAZARD SUMMARY

STATION	PRIMARY HAZARD	WHY IT'S DANGEROUS
01 Inspect/Load	Sharp/heavy parts; pinch points	Cuts, crush, caught-in
02 Clean/Degrease	Hot caustic or flammable/inhalable solvent	Chemical burns; eye injury; fire; inhalation
03 Rinse	Residual cleaner; wet floor	Mild irritant; slips
04 Passivate	Nitric: oxidizer/corrosive + NOx fumes ; Cr(VI) if dichromate; hot acid	Severe burns; toxic fumes; carcinogen ; scalds
05 Rinse	Acid residue; wet floor	Burns/irritation; slips
06 Dry	Heat (if oven); hot parts	Burns; heat stress
07 Test	Test reagents (copper sulfate, ferricyanide, acids)	Irritants/toxics — handle per SDS

⚠

HEADS-UP
Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting process chemicals — and several test reagents (**potassium ferricyanide, copper sulfate**) are toxic. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

CHAPTER 6

EMERGENCY & FIRST-RESPONSE

KNOW THESE BEFORE YOUR FIRST SHIFT, NOT DURING YOUR FIRST INCIDENT



HEADS-UP

Memorize the location of the nearest **eyewash**, **safety shower**, **fire extinguisher**, **spill kit**, **first-aid kit**, and **emergency exit** *before* you start work. In an emergency you won't have time to go looking.

FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Acid on skin (nitric or citric)	Remove contaminated clothing, flush with water at least 15 minutes , get help. Don't "wipe and continue."
Acid in eyes	Go straight to the eyewash, hold eyelids open, flush at least 15 minutes , get medical help. Eyes don't get second chances.
Inhaled NOx / acid fumes (cough, throat/chest tightness, shortness of breath)	Get to fresh air immediately and get medical help — even if you feel okay. NOx can cause delayed, serious lung injury hours later. Report it; ventilation may be failing.
Degreaser on skin/eyes	Flush 15 minutes; for solvents, also move to fresh air. Get help.
Burn from a hot bath or hot part	Cool with water; do not peel clothing stuck to a burn; get medical help for anything beyond minor.
Spill	Only clean up small spills you're trained for, in PPE. Never mix nitric acid with organics, solvents, or rags (violent reaction, fire). Acid and alkaline cleaner are opposites — never just mix them to "cancel out." Follow your shop's spill procedure.
Any Cr(VI) / dichromate exposure	Treat as serious; flush, report, follow the specific SDS and your shop's Cr(VI) exposure procedure.



REMEMBER

When in doubt, **flush, then ask**. Water and time fix most chemical contact if you act fast. Reporting every incident — even "minor" ones, and *especially* any fume exposure — is how the shop fixes the cause before it hurts the next person.



HEADS-UP — LOCKOUT/TAGOUT (LOTO)

Heaters, pumps, hoists, conveyors, and dryers are powered by serious electrical equipment. **LOTO is mandatory before any maintenance** — physically lock the power off and tag it so no one can switch it on while you're working.



FUN FACT

The 15-minute flush rule isn't arbitrary — it's the time emergency-response standards found necessary to dilute and carry away most corrosive chemistry before it does deep tissue damage. That's why eyewash and shower stations must deliver a steady, hands-free flow for a full 15 minutes.

CHAPTER 7

THE SAFETY PHILOSOPHY OF A PASSIVATION LINE

“NO COATING” DOES NOT MEAN “NO HAZARD” — NITRIC IS THE ONE TO RESPECT

People sometimes assume that because passivation “just cleans” stainless and “doesn’t add anything,” it’s harmless. That’s a dangerous assumption. The hazards are real — they’re just concentrated in a few places:

- **Nitric acid is the headline hazard.** Strong oxidizer, severe burns, and — uniquely — it can throw off **toxic brown NO_x fumes**, especially during flash attack. NO_x is the sneaky one: it can feel mild and injure your lungs hours later. **Ventilation is a primary control, not an afterthought.**
- **Hexavalent chromium, if dichromate is used.** A carcinogen with its own full OSHA program. The cleanest fix is a chemistry that doesn’t use it (plain nitric or citric).
- **Hot acid baths.** Many passivation baths run warm to hot (up to ~160°F / 71°C) — scald and splash hazards.
- **The degreaser.** Either hot caustic (burns) or solvent (fire/inhalation), depending on your shop.
- **Test reagents are toxic too.** Copper sulfate and potassium ferricyanide are not “just lab stuff” — handle per SDS.



HEADS-UP

Citric passivation exists in large part to remove the worst of these hazards. It’s biodegradable, far less corrosive, produces **no NO_x**, and uses **no Cr(VI)**. When a shop has the choice (and the customer spec allows it), citric is usually the safer, greener call. That’s why it’s the growing industry standard. (Final chemistry decisions go through your QC/EHS process — this manual advises, it doesn’t approve.)



REMEMBER

The safety philosophy of this line is simple: **respect the nitric, control the fumes, know whether dichromate is in your bath, and handle the test reagents like the chemicals they are.** A passivation line isn’t “the safe line” because nothing’s deposited — it’s a line built around **nitric acid**, which deserves the same discipline you’d give any strong acid.



REMEMBER

You’ve now got the whole foundation: what passivation is (and why it’s not a coating), the role of free iron and the chromium-oxide passive layer, nitric versus citric, how the seven-station line works as one connected system, why the order is locked by chemistry, what to wear, what to do in an emergency, and the safety mindset that ties it together. Everything in the station chapters builds on these fundamentals. Turn the page knowing the *why* — and the *how* will make a lot more sense.

PART TWO — THE LINE, STATION BY STATION (STATIONS 01–07)
STATION 01 OF 07

INSPECT & LOAD

“GARBAGE IN, GARBAGE OUT — AND ON STAINLESS, THE WRONG PREP MEANS THE ACID CAN’T HELP YOU.”

Before any chemistry touches a part, someone has to confirm it’s the right part, in the right condition and the right alloy, and load it so the acid can reach every surface. It feels like the un-glamorous step. It isn’t: a part racked wrong gets cleaned wrong, passivated wrong, and tested wrong — and on passivation, where there’s no coating to hide behind, those misses show up as rust in the field.

• WHAT YOU’RE DOING & WHY IT MATTERS

- **Confirm the job and the alloy.** Right part number, quantity, spec — and critically, **which stainless grade**. Grade drives everything: a forgiving 304 austenitic and a flash-attack-prone 416 free-machining or 440C martensitic part need *different passivation conditions* (see Part Three). Check the traveler for the **governing spec** (ASTM A967 / AMS 2700 type, or customer spec) and the **required test**.
- **Inspect the incoming condition.** Heavy weld scale, heat tint (dark discoloration near welds), or mill scale is a **STOP** — passivation will *not* remove it. Those parts need **pickling or mechanical descaling first**. Heavy grease or dried coolant means extra cleaning attention.
- **Load for full contact and full drainage.** Every surface must see the acid, and every cavity must **drain** — trapped acid in a cup, channel, or blind hole becomes a stain pocket, a flash-rust spot, and a safety hazard when handled.

• OPERATOR ESSENTIALS

ITEM	GUIDANCE
Alloy/grade check	Confirm grade; flag free-machining (303, 416) & martensitic/high-carbon (410, 420, 440) → gentler conditions
Spec check	Confirm A967/AMS 2700 type & required verification test from traveler
Surface condition	Weld scale / heat tint / mill scale → STOP, route to pickling or descale first
Orientation	Rack so liquids drain freely — no cups, pockets, or trapped air
Spacing	Parts not touching/nesting — acid and rinse must reach all faces
Mixed-metal flag	No carbon-steel parts/fixtures in the load — they contaminate the bath with iron

**HEADS-UP**

Mechanical hazards rule this station. Sharp edges, heavy parts, pinch points at racks and hoists. Wear cut-resistant gloves for handling, lift with your legs, get help for heavy loads. Also: **never rack stainless on carbon-steel fixtures or with carbon-steel wire** — you’d be adding the very free iron you’re trying to remove.



TIP

Use **stainless, titanium, or plastic** racks, baskets, and wire for stainless parts — never carbon steel. Cross-contamination at the rack is a classic, invisible cause of “why won’t this part pass the copper sulfate test?”

• **STEP-BY-STEP PROCEDURE**

1. **Read the traveler.** Confirm part, quantity, **grade**, governing spec, and required test.
2. **Inspect each part** for weld scale, heat tint, mill scale, heavy grease, or damage. Route anything with scale/tint to pickling or descaling — *not* into passivation.
3. **Rack or basket** so every surface is exposed and every cavity drains. Use only stainless/titanium/plastic fixtures.
4. **Confirm spacing and balance** so parts don’t nest or fall.
5. **Note anything unusual** (mixed grades, sensitive alloys, scale) to your lead before the load enters the cleaner.

• **TOP MISTAKES NEW OPERATORS MAKE**

1. **Loading scaled or heat-tinted parts** expecting passivation to clean them — it can’t (that’s pickling).
2. **Nesting parts** so faces touch — those faces don’t get cleaned or passivated.
3. **Not flagging free-machining/martensitic grades** — they may flash-attack in standard nitric.
4. **Using carbon-steel racks/wire** — you contaminate the part and the bath with free iron.
5. **Racking so cavities trap acid** — guaranteed stains and a hazard at unload.

• **TEACH-BACK CHECKLIST**

- ☐ Explain why drainage orientation matters (trapped acid = stains, flash rust, hazard).
- ☐ State which incoming conditions are a STOP (weld scale, heat tint, mill scale → pickle/descale first).
- ☐ Explain why fixtures must be stainless/titanium/plastic, never carbon steel.
- ☐ Identify which grades need flagging for gentler conditions (free-machining, martensitic/high-carbon).

SIGN-OFF — STATION 01

Trainee: _____ Trainer: _____ Date: _____

“I confirm I verified the job, grade, and spec; inspected for scale/heat tint; loaded parts on non-ferrous fixtures for full contact and free drainage; and handled parts safely in PPE.”

STATION 02 OF 07 • CRITICAL

CLEAN / DEGREASE

“PASSIVATE OVER OIL AND YOU’VE PASSIVATED NOTHING.”

This is where oil, grease, fingerprints, coolant, and shop soil come off so the stainless is bare and the acid can actually reach the surface. **On a passivation line this is the single most important preparation step** — and the most common cause of failure when skipped or shortchanged. The passivating acid cannot dissolve free iron it can’t touch, and oil blocks it completely. **Clean stainless is non-negotiable.**

• WHAT YOU’RE DOING & WHY IT MATTERS

The cleaner removes **all** organic soil — oils, greases, drawing/cutting fluids, fingerprints, waxes, and particulate — leaving a uniformly clean, water-loving surface that passes the **water-break test**. Two common cleaner families:

- **Alkaline (high-pH) hot detergent** — emulsifies and lifts oils; very common for passivation prep.
- **Solvent or vapor degrease** — dissolves oils; used where alkaline isn’t suitable.

 **REMEMBER**
Every other station depends on this one. The acid bath, the rinse, the test — all of it is wasted effort if the part walked in oily. When troubleshooting *any* passivation failure, the first question is always: *was it truly clean?*

 **TECHNICAL STUFF**
Alkaline cleaners for stainless are formulated to degrease without aggressively etching the surface; builders/buffers, surfactants, and sometimes mild chelators do the work. Some shops use a dedicated **alkaline derust/descale** step or a **citric pre-clean** to lift light oxides — but remember none of these remove *heavy* scale (that’s pickling). Exact concentration, temperature, and time come from the **supplier TDS** — never guess.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	120–180°F (49–82°C) hot alkaline	Per TDS; don’t exceed max
Time	5–15 min (or per TDS)	Heavier soil = longer
Concentration	Per supplier TDS (titrate)	Too weak won’t cut oil
pH (alkaline)	~10–13 per product	Caustic — burns
Agitation	Per equipment	Keeps fresh cleaner on the surface
Oil control	Skim floating oil	Floating oil redeposits on parts

 **HEADS-UP**
If hot alkaline: this tank is **hot and caustic** — splashes cause **chemical burns**, and steam/mist irritates airways. If solvent: **flammable and/or an inhalation hazard** — control ignition sources and ventilate. **PPE every time:** chemical splash goggles, face shield (charging/dumping), elbow-length gloves, apron, chemical-resistant boots. First aid: skin → flush 15 min; eyes → eyewash 15 min and get help.

• STEP-BY-STEP PROCEDURE

- 1. **Check the bath before you start.** Confirm temperature and concentration are in spec (last titration log). Skim surface oil.
- 2. **Inspect the load.** Heavy oil → plan the long end of the time range.
- 3. **Clean fully** for the full time, with agitation reaching all faces.
- 4. **Withdraw and drain** back into the tank (if dip).
- 5. **Run the water-break test** on a sample part: continuous sheet = pass; beading = fail, re-clean.
- 6. **Move promptly** to the rinse so the clean surface doesn't sit and pick up airborne soil.



TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)

A truly clean stainless surface loves water, so it **sheets off in one continuous, unbroken film**. A dirty surface repels water, so it **beads up** into droplets. **Continuous sheet = PASS** (send it on). **Beading = FAIL** (re-clean). On passivation this isn't a nicety — it's the difference between the acid working and not working at all.

• TOP MISTAKES NEW OPERATORS MAKE

- 1. **Skipping the water-break test** because the part “looks clean.” The oil that ruins passivation is invisible.
- 2. **Running the bath too cold or too weak** — it can't cut oil no matter how long you soak.
- 3. **Letting surface oil build up** with no skimming — it redeposits on the way out.
- 4. **Nesting parts** so cleaner can't reach every face.
- 5. **Touching cleaned parts bare-handed** — fingerprints are oil; they cause local passivation failures and copper-sulfate-test spots.

• QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Water-break fails	Low concentration or temperature	Titrate; bring temp to range
Oily film after cleaning	Floating oil redepositing	Skim; refresh/dump if heavily loaded
Part fails passivation test later	Inadequate clean upstream	Re-clean; verify water-break before acid
Fingerprint-shaped rust spots	Bare-handed handling after clean	Handle with clean gloves only

SIGN-OFF — STATION 02

Trainee: _____ Trainer: _____ Conc./Temp: _____

“I confirm parts passed the water-break test, the cleaner was in concentration and temperature range, parts were handled with clean gloves, and I wore required PPE.”

STATION 03 OF 07

RINSE (POST-CLEAN)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT.”

You just got the stainless clean — but it’s coated in a film of degreaser. This station’s whole job is to **wash that film off** before the part hits the acid stage. It looks like a “nothing” step. It isn’t: a weak rinse here drags alkaline cleaner into the acid bath, neutralizing and weakening it.

• WHAT YOU’RE DOING & WHY IT MATTERS

An alkaline cleaner is high-pH; the passivating bath is acidic. Carry cleaner forward and it **neutralizes acid and drifts the bath out of spec**. The invisible carry-over of solution clinging to a part is **drag-out** (leaving a tank) / **drag-in** (arriving at the next). The rinse knocks drag-out down to almost nothing by dilution: clean water surrounds the part, the cleaner film dissolves into the rinse, and constant flow flushes it away.

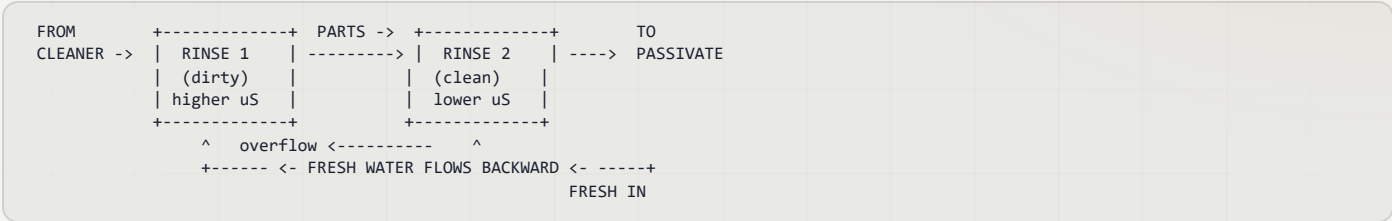


TIP — HOW WE MEASURE A CLEAN RINSE: CONDUCTIVITY

You can’t see dissolved cleaner, so we measure **conductivity** — how well the water conducts electricity. Pure water barely conducts; dissolved chemicals make it conduct more. So **higher conductivity = more cleaner left in the rinse = dirtier rinse**. It’s measured in **microsiemens per centimeter (µS/cm)** — just think “the contamination number.” Lower is cleaner.

• THE SMART WAY TO RINSE: COUNTER-FLOW

The professional approach is a **counter-flow (counter-current) rinse**: parts move forward, water flows backward. Fresh water enters the *last* (cleanest) stage and overflows backward into the dirtier one, so the cleanest water the part ever sees is the last thing to touch it before the acid.



HEADS-UP

Rinse water carries dragged-in cleaner. Wear **goggles, chemical gloves, and non-slip footwear** — the floor around rinses is always wet, and **slips are the #1 injury here**.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (60–85°F)	No heating required
Time	1–2 min with agitation	Longer for blind holes/threads
Water source	Municipal or DI	DI rinses cleaner, fewer spots
Conductivity	Keep low (e.g. < 500 µS/cm)	Monitor; rising = drag-in
Flow	Continuous overflow / counter-current	Saves water

• STEP-BY-STEP PROCEDURE

1. **Check conductivity at shift start.** Rising/high → tell your lead; the rinse needs more flow or a refresh.
2. **Confirm water is flowing** (overflow or counter-current active).
3. **Transfer promptly** from the cleaner — dried cleaner is harder to rinse off.
4. **Rinse fully, 1–2 min**, with agitation; flush blind holes and threads longer.
5. **Drain** briefly to carry less forward, then move to the acid stage.

• TOP MISTAKES NEW OPERATORS MAKE

1. Treating the rinse as “quick” — a two-second dunk rinses nothing.
2. Ignoring conductivity — a high number means you’re blunting the acid right now.
3. Not flushing blind holes/threads — concentrated cleaner hides and rides forward.

• TEACH-BACK CHECKLIST

- ☐ Explain why carrying cleaner into the acid causes problems (neutralizes/weakens it).
- ☐ Define drag-out / drag-in.
- ☐ Explain what conductivity tells you and which direction is “clean” (lower = cleaner).
- ☐ State the rinse time and why threads/blind holes need longer.

SIGN-OFF — STATION 03

Trainee: _____ Trainer: _____ Conductivity: _____ µS/cm

“I confirm rinse conductivity was within limit, water flow was running, and parts were rinsed the full time with PPE worn.”

STATION 04 OF 07 • THE MAIN EVENT

PASSIVATE (NITRIC OR CITRIC)

“THIS IS THE STATION WHERE FREE IRON COMES OFF AND THE STAINLESS’S OWN ARMOR GETS STRONGER.”

Now — with a clean, bare, well-rinsed stainless surface — the passivation actually happens. The acid dissolves free iron and contamination and helps the chromium-oxide passive layer reform stronger, thicker, and more uniform (Chapters 1–2 covered the chemistry). Everything before this stage existed to deliver a perfect surface here; everything after exists to protect and *prove* what happens here.



REMEMBER — THE TWO CHEMISTRIES

This stage is either a **nitric** bath (classic; sometimes with sodium dichromate for tough grades) or a **citric** bath (modern, safer, greener). Same goal — remove free iron, enhance the passive film — different hazards and different parameters. **Know which one you’re running before you start.**

• WHAT YOU’RE DOING & WHY IT MATTERS

The acid does two jobs at once: it **dissolves free iron and surface contamination**, and it **enriches/rebuilds the chromium-oxide passive layer** (by leaving the surface chromium-rich and helping oxygen reform the film). **No coating is deposited** — the part comes out looking essentially the same as it went in, only cleaner. The proof is in the test (Station 07), not in the appearance.



TECHNICAL STUFF — BATH CONTROL

Passivation baths are controlled by **concentration** (nitric: % by volume; citric: % by weight), **temperature**, and **time** — all defined by the governing spec’s *type/method* for your grade. Nitric baths are checked by titration/specific gravity; citric by concentration and (importantly) **microbiological/cleanliness control** since citric can grow microbes. The exact “type” (e.g., the specific Nitric or Citric treatment number in ASTM A967, or the Method/Type in AMS 2700) is dictated by your customer’s spec and the alloy — treat the numbers as *representative*; *verify against your spec and TDS.*

• OPERATOR ESSENTIALS

PARAMETER	NITRIC	CITRIC	NOTES
Concentration	~20–50% by vol HNO ₃	~4–10% by wt citric	Per spec type/TDS
Additive (some grades)	Sodium dichromate (Cr(VI)!)	Passivators/wetting (no Cr)	Dichromate only on certain nitric types
Temperature	70–160°F (21–71°C)	~120–160°F (49–71°C)	Higher temp = faster, more flash risk
Time	~20–30+ min	~4–30 min	Per spec; longer for tougher cases
Hazard headline	Oxidizer + NO _x ; Cr(VI) if dichromate	Mild acid; no NO _x /Cr(VI)	Drives PPE/ventilation

HEADS-UP — NITRIC ACID + NOX + CR(VI)

If this is a **nitric** bath:

- It's a **strong oxidizer and corrosive** — severe burns, eye damage.
- Watch for **brown/red-brown NOx fumes**, especially if the bath is attacking the metal (flash attack). **NOx is a serious inhalation hazard with delayed lung effects** — local exhaust ventilation is required; respirator if ventilation is inadequate.
- **Never let nitric contact organics, solvents, or rags** — violent reaction/fire.
- If it contains **sodium dichromate**, the full **OSHA Cr(VI) program (29 CFR 1910.1026)** applies.

PPE: goggles + face shield, nitric-rated gloves, apron, boots, respirator as required.

HEADS-UP — CITRIC

Much milder and far safer, **but still an acid** and often **hot** — splash and scald hazards remain. No NOx, no Cr(VI). Wear goggles, gloves, apron, boots.

• **READING THE RESULT (SPOILER: THERE'S LITTLE TO SEE)**

Unlike a coating, a properly passivated part looks almost identical to how it went in — clean, bright stainless. There is no color, no film thickness to read. What you *can* watch for is trouble:

WHAT YOU SEE	MEANING	ACTION
Bright, clean, unchanged surface	Normal — passivation adds no visible coating	Proceed to rinse; confirm by test
Darkening, frosting, etching, or a rainbow/rusty haze	Flash attack — acid attacked the steel itself	Stop; this is a reject; review conc/temp/grade
Brown fumes off the bath	NOx — over-aggressive reaction / flash attack	Ensure ventilation; investigate bath & grade
Smut or dark film	Contamination or wrong conditions for grade	Re-evaluate; may need re-clean/re-process

REMEMBER

A good passivation job is boring to look at. The part comes out looking like clean stainless — because that's all it is. If the surface comes out *darkened, frosted, etched, or hazy*, that's not "extra passivation," it's **flash attack** — a defect (covered in depth in Part Three). With passivation, "nothing visibly changed" is the goal; **the test is your confirmation.**

• STEP-BY-STEP PROCEDURE

- 1. **Confirm which chemistry** (nitric vs citric) and the matching PPE/ventilation (full Cr(VI) program if dichromate).
- 2. **Confirm bath health:** concentration, temperature, and (citric) microbiological/cleanliness control — all per the spec type for this grade.
- 3. **Confirm the part arrived clean and well-rinsed** (Stations 02–03). A poor clean here = failed passivation.
- 4. **Immerse for the correct time** (typically 20–30+ min nitric; per spec for citric) at the correct temperature.
- 5. **Watch for flash attack / brown fumes** — stop and report if you see darkening, etching, or NOx.
- 6. **Withdraw and move promptly to rinse** — don't let acid dry on the part.
- 7. **Log** bath checks and any flash-attack/fume concerns to your lead.

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Part still rusts / fails test	Bath too weak/cold/short; or part wasn't clean	Verify conc/temp/time; re-check cleaning
Darkened / etched / frosted surface (flash attack)	Acid too strong/hot for grade; free-machining/martensitic alloy; chloride contamination	Use gentler type (lower temp, dichromate, or citric); check for chlorides
Brown fumes	Over-aggressive reaction / flash attack (nitric)	Ventilate; investigate; consider gentler conditions
Stains/streaks after	Acid dried on part; bad rinse	Rinse promptly and thoroughly; use DI final
Citric bath cloudy/smelly	Microbial growth	Treat/refresh per TDS; maintain bath control

• TEACH-BACK CHECKLIST

- ☐ Explain what passivation does — removes free iron, enhances the chromium-oxide passive layer, **adds no coating**.
- ☐ State that a good result looks unchanged, and that **darkening/etching = flash attack = reject**.
- ☐ Name the two chemistries and the headline hazard of each (nitric: NOx/Cr(VI); citric: mild but hot acid).
- ☐ Explain why you confirm success by **testing**, not by looking.
- ☐ State what brown fumes mean and the required response (ventilation, report).



TIP

Passivation rewards patience. The most common “the part failed the test” cause that isn’t cleaning is simply **not enough time at temperature**. Resist the urge to pull baskets early to keep the line moving — a part that spent 8 minutes in a bath that needed 25 looks done and isn’t.



TECHNICAL STUFF — WHY CITRIC AND NITRIC BOTH COUNT AS “PASSIVATION”

It's natural to assume a stronger acid must passivate “harder.” It doesn't work that way. **Nitric** dissolves free iron *and* actively oxidizes/thickens the chromium film. **Citric** dissolves free iron and then *chelates* (grabs and holds) the iron ions in solution so they can't redeposit, while dissolved oxygen reforms the passive layer. Different routes, same destination: less surface iron, a better native passive film. The governing specs treat them as **equivalent-when-qualified** methods — success is proven by the *verification test*, not by which acid sat in the tank.



REMEMBER

The passivate stage is where the whole line pays off — but it cannot rescue a bad upstream. If the part arrived oily (Station 02 failed) or carrying alkaline drag-in (Station 03 failed), the best bath in the shop won't passivate it. When a load fails the test, walk it *backward*: clean first, then rinse, then bath chemistry, then grade/conditions.



HEADS-UP — BATH ADDITIONS

Adjusting a passivation bath — especially adding acid or **sodium dichromate** to a nitric tank — is done only by authorized personnel, in full PPE, with ventilation running. Adding to a hot oxidizing acid can drive sudden reaction and **NOx**. Never “top off” a bath on your own judgment; follow the posted procedure and the TDS.

SIGN-OFF — STATION 04

Trainee: _____ Trainer: _____ Chem/Conc/Temp: _____

“I confirm I identified the chemistry, verified concentration/temperature/time per the spec for the grade, watched for flash attack and fumes, ran ventilation, moved parts promptly to rinse, and wore required PPE (including the full Cr(VI) program if dichromate).”

STATION 05 OF 07

RINSE (POST-PASSIVATE)

“STOP THE REACTION, WASH AWAY THE IRON, AND DON’T LET ACID DRY ON STAINLESS.”

The freshly passivated part leaves the acid wet with spent acid and dissolved iron. This rinse stops the reaction, washes that residue away, and prevents acid from drying on the surface — which would stain the part and could even re-attack it. Ideally the *last* rinse uses **clean or deionized (DI) water** so no minerals spot the surface.

• WHAT YOU’RE DOING & WHY IT MATTERS

Two jobs: **stop the passivation reaction** by washing away the acid, and **deliver a clean, residue-free surface** to drying and testing. Leftover acid or dissolved-iron residue can **stain, streak**, or cause **flash rust** — and it will throw off your verification test.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (some specs: warm final rinse)	Per spec
Time	1–2 min with agitation	Longer for threads/cavities
Water	Final rinse: DI preferred	Low residue = no spots
Conductivity	Low (per shop)	Rising = acid drag-in
Flow	Continuous / counter-current	Same logic as Station 03

**HEADS-UP**

Acid residue and wet floors. Goggles, gloves, non-slip footwear. **Don’t let rinsed parts sit wet** — pooled water can spot, and acid trapped in a cavity can keep working.

• STEP-BY-STEP PROCEDURE

1. **Confirm flow and conductivity** at shift start.
2. **Rinse fully, 1–2 min**, flushing cavities and threads.
3. **Use the cleanest/DI water last.**
4. **Drain briefly**, then move promptly to dry — no sitting, no spotting.



REMEMBER

Acid left to dry on stainless stains it — and can undo your work. This rinse is the wall between a clean, passivated surface and a stained, re-contaminated one. The final rinse before drying should be your cleanest water (ideally DI), because whatever is on the surface now is what the test will read.

• TOP MISTAKES & TEACH-BACK

TOP MISTAKES TO AVOID

- Skimping on rinse → acid residue → stains and failed tests.
- Using hard/dirty water for the *final* rinse → spots and residue.
- Letting parts sit wet → water spots / flash rust.

TEACH-BACK — CAN YOU ANSWER?

- What this rinse removes and why residue stains/fails tests?
- Why DI water is preferred for the final rinse?
- The flash-rust / spotting risk of letting parts sit wet?



TIP

If your finished parts keep coming out spotty even though the chemistry checks out, your rinse water is the prime suspect. Hard, chloride-laden, or stagnant rinse water spots stainless and can even feed flash attack — a clean DI final rinse fixes more “mystery” cosmetic rejects than any bath adjustment.

SIGN-OFF — STATION 05

Trainee: _____ Trainer: _____ Conductivity: _____ μ S/cm

“I confirm the post-passivate rinse was flowing and in range, the final rinse was clean/DI, and parts moved promptly to drying with PPE worn.”

STATION 06 OF 07

DRY

“A CLEAN PART THAT AIR-DRIES WITH HARD-WATER SPOTS LOOKS LIKE A DEFECT.”

Drying removes all water so the part comes out clean, dry, and spot-free, and ready for handling and testing. Water left to evaporate leaves mineral spots and can even start surface staining.

• WHAT YOU'RE DOING & WHY IT MATTERS

Remove **all** surface and trapped water — by warm forced air, a low-temperature oven, or a clean wipe/blow-off — before handling and testing. A DI final rinse plus prompt drying gives a spotless surface. (Note: drying after passivation is gentle — there's no coating to cure, just water to remove.)

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Method	Warm air, oven, or clean blow-off	Per shop/part
Temperature	Ambient–250°F (avoid high heat that could heat-tint thin sections)	Per part
Goal	Bone dry, including cavities/threads	No water spots



HEADS-UP

If an oven or heated air is used, watch for **burn and heat-stress** hazards; use heat-resistant gloves for hot parts. Don't over-heat thin stainless to the point of discoloration.



REMEMBER

A passivated part can be perfect chemically and still *look* bad if it air-dries with water droplets. Proper drying (after a DI rinse) is what makes the finish look as clean as it is.

• STEP-BY-STEP PROCEDURE

1. **Confirm drying method/temperature** is set and appropriate for the part.
2. **Dry fully**, including cavities and threads (no droplets, no steaming).
3. **Handle with clean gloves** — bare-handed handling re-deposits oils/iron and ruins the just-finished surface.
4. **Move to test/verify** while clean and dry.

• TOP MISTAKES & TEACH-BACK

TOP MISTAKES TO AVOID

- Letting parts air-dry → water spots.
- Bare-handing dried parts → fingerprints/free iron → failed copper-sulfate test.
- Overheating thin sections → heat tint.

TEACH-BACK — CAN YOU ANSWER?

- Why parts shouldn't air-dry (spots/staining)?
- Why dried parts must be handled with clean gloves?
- Why not over-heat thin stainless (heat tint)?

SIGN-OFF — STATION 06

Trainee: _____ Trainer: _____ Date: _____

"I confirm parts were dried fully (including cavities), handled with clean gloves, and not overheated."

TEST & VERIFY

“WITH PASSIVATION, THE TEST ISN’T PAPERWORK — IT’S THE ONLY WAY ANYONE KNOWS IT WORKED.”

This is the station that makes passivation different from every coating process. There is **no film to inspect and no color to read**. A perfectly passivated part and a completely un-passivated part can look *identical*. The only way to know the part is properly passivated — and the way the customer’s spec defines acceptance — is to **test it**.

• WHAT YOU’RE DOING & WHY IT MATTERS

You run the **verification test named in the governing spec** (ASTM A967 / AMS 2700 / customer spec) on the parts or on test samples processed with them. The test confirms the surface is **free of free iron** and **adequately passive**. Passing = the job is good. Failing = re-clean and re-process (or investigate). The most common tests:

TEST	WHAT IT DETECTS	PASS / FAIL LOOK
Water immersion	Free iron / poor passivity	Rust/staining after cycles = fail
High humidity	Free iron / poor passivity	Rust spots in humidity cabinet = fail
Salt spray (ASTM B117)	Overall corrosion resistance	Rust within spec’d hours = fail
Copper sulfate	Free iron (fast, common)	Copper-colored (pink) deposit = free iron = fail
Ferroxyl / K-ferricyanide-nitric	Free iron (very sensitive)	Blue spots = free iron = fail

★

REMEMBER

With a coating you can see the result; with passivation you test for it. This station is the proof of the entire line. A part that *looks* perfect can still be full of free iron — only the test tells the truth. Treat the test station as seriously as the acid bath.

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HEADS-UP — TEST REAGENTS ARE TOXIC

Copper sulfate and potassium ferricyanide are hazardous chemicals — handle per their SDS, in PPE, with proper disposal. Potassium ferricyanide must **never** be mixed with strong acid in a way that could release cyanide gas — follow the prepared-reagent procedure exactly. Don’t improvise test chemistry.

**TIP — GRADE CAUTION ON COPPER SULFATE**

The **copper sulfate test is not appropriate for all grades** — it can give false readings on free-machining or lower-chromium (martensitic, e.g., some 400-series) stainless, and it's generally **not used on parts that contact food or that can't tolerate copper contact**. Always use the test method *specified for that grade and application* — the spec tells you which test is valid.

• OPERATOR ESSENTIALS

- **Run the test the spec names** — don't substitute a faster test for the required one.
- **Test clean, dry parts** — residue or moisture gives false results.
- **Use control samples** where required, processed with the load.
- **Log results** against the part/lot and the spec acceptance criteria.

• STEP-BY-STEP PROCEDURE

1. **Confirm which test** the traveler/spec requires for this grade and application.
2. **Prepare reagents/equipment** per procedure and SDS, in PPE.
3. **Test the parts or samples**, observing for the specified time/cycles.
4. **Read the result against the spec's acceptance criteria** (e.g., no copper deposit / no blue / no rust within X hours).
5. **Pass → release; Fail → quarantine, re-clean, re-process, and re-test**. Notify your lead on any failure.
6. **Record** results and dispose of test reagents per the waste procedure.

• TOP MISTAKES & TEACH-BACK**TOP MISTAKES TO AVOID**

- Eyeballing instead of running the required test (you can't see passivation).
- Using the wrong test for the grade (false pass/fail).
- Testing wet/dirty parts (false results).
- Mishandling toxic reagents.

TEACH-BACK — CAN YOU ANSWER?

- Why passivation must be confirmed by testing, not appearance?
- What a copper-sulfate "pink" and a ferroxyl "blue" each mean (free iron = fail)?
- Why copper sulfate isn't valid for every grade?
- The hazards of the test reagents?

SIGN-OFF — STATION 07

Trainee: _____ Trainer: _____ Test/Result: _____

"I confirm I ran the spec-required verification test on clean, dry parts/samples, read it against the acceptance criteria, recorded the result, quarantined any failures, and handled test reagents safely per SDS."

• WHY THE LINE ENDS HERE — AND NOT AT A VISUAL CHECK

On every coating line in our other manuals, the last station is an *inspection*: you look at the color, the weight, the finish, and you can tell at a glance whether the job is right. Passivation breaks that habit on purpose. Because **nothing was deposited**, there is nothing to look at — so the line ends with a *test* instead of a look. The verification test is the passivation equivalent of reading a coating's color: it's how the operator, the inspector, and the customer all agree the work is good.



REMEMBER

Two checks bracket every passivation job: the **water-break test** at the front (proves the part was clean enough for the acid to work) and the **verification test** at the back (proves the free iron is gone and the surface is passive). Front and back, clean and proven — that's what makes an invisible process trustworthy.



FUN FACT

The ferroxyl (potassium-ferricyanide) test turns free iron a vivid **Prussian blue** — the same pigment used by artists and in old “blueprint” drawings. When an inspector dabs the reagent on a part and a blue freckle blooms, they're literally watching a 300-year-old pigment reaction flag a single speck of stray iron.



TIP

A copper-sulfate or ferroxyl free-iron test takes only minutes. Many shops run a quick free-iron check as an in-process gate on every lot and reserve the slow tests (salt spray, humidity) for periodic qualification per the spec — catching free iron at the bench instead of in the customer's hands.

PART THREE — ACROSS THE LINE
— SAME GOAL, TWO CHEMISTRIES

NITRIC VS. CITRIC : THE DECISION

SAME GOAL, TWO VERY DIFFERENT CHEMISTRIES — AND THE INDUSTRY IS SHIFTING

	NITRIC ACID	CITRIC ACID
Status	Classic, long-established reference	Modern, increasingly the standard
Removes free iron	Yes	Yes
Enhances passive layer	Yes (strong oxidizer)	Yes (chelates iron; film reforms)
NOx fumes	Yes — toxic brown fumes, esp. flash attack	None
Cr(VI)	Only if sodium dichromate added (carcinogen)	None
Flash-attack risk	Higher on sensitive grades (mitigated by dichromate/temp control)	Lower / gentler
Environmental	Nitrate (and chromium) in waste — regulated	Biodegradable , easier waste
Safety	Strong oxidizer/corrosive	Much milder (still an acid, often hot)
Common limitation	Hazard handling, ventilation, waste	Microbial bath maintenance; some specs still require nitric

★

REMEMBER

Nitric is the classic but hazardous choice (NOx; Cr(VI) if dichromate). Citric is the modern, safer, greener choice (biodegradable, no NOx, no Cr(VI)) and is increasingly the industry standard. The right answer is driven by the customer's spec (which may require a specific method), the alloy (sensitive grades favor citric or gentler nitric types), and your shop's safety/environmental priorities.

• HOW THE DECISION GETS MADE (IN PRACTICE)

You usually won't choose the chemistry yourself, but you should understand what drives the choice:

- **The customer's spec comes first.** If the drawing/traveler calls out a specific method or type, run *that*. Some specs let you pick nitric or citric; some demand one.
- **The alloy matters.** Forgiving austenitic 300-series runs fine either way; flash-attack-prone free-machining and martensitic grades favor citric or a gentler, dichromate-inhibited nitric type.
- **Safety and environment tip the scale toward citric** when there's a free choice — no NO_x, no Cr(VI), biodegradable waste.



TECHNICAL STUFF

Both ASTM A967 and AMS 2700 treat nitric and citric as legitimate methods, defined by **types/methods** that pin down concentration, temperature, time, and (for some nitric types) dichromate addition by alloy family. Acceptance is proven by the **verification test**, not by which acid you used — so a citric process that passes the spec's required test is just as "passivated" as a nitric one. When in doubt whether you may substitute citric for a nitric callout, **get written approval against the governing spec** before changing.



HEADS-UP

Never swap nitric for citric (or change a type) on your own judgment to "make the line safer" or "go faster." The method is part of the customer's qualified process. Chemistry changes go through QC/EHS with written approval — this manual advises, it doesn't approve.



FUN FACT

The "greener" passivation chemistry is literally the same acid that makes lemons and oranges sour. Citric acid passivation lets a shop replace a tank of fuming nitric with something biodegradable — one of the rare cases where the safer, more environmentally friendly option also works *better* on sensitive alloys.

— NOT ALL STAINLESS IS EQUAL

GRADE-DEPENDENCE & FLASH ATTACK

WHY THE SAME BATH THAT'S PERFECT FOR 304 CAN RUIN A 416

Not all stainless is equally easy to passivate. The alloy's **chromium content, carbon content, and added elements (like sulfur for machinability)** change how it reacts with the acid.

- **Austenitic 300-series (304, 316, etc.): forgiving.** High chromium, low carbon — passivate readily in standard nitric or citric.
- **Free-machining grades (303, 416): touchy.** Added **sulfur** (manganese sulfide inclusions) makes them machine beautifully but react aggressively with acid — prone to **flash attack**.
- **Martensitic / high-carbon (410, 420, 440C): touchy.** Higher carbon and lower effective chromium make them more reactive and more easily over-etched.

• WHAT IS FLASH ATTACK?

Flash attack (also “flash etching”) is when the passivating acid stops politely dissolving free iron and starts **aggressively attacking the stainless itself**. Instead of coming out bright and clean, the part comes out **darkened, frosted, etched, hazed, or even rusty** — and with nitric, it often drives off **brown NO_x fumes**. It's a **defect** — the opposite of what you want. Common causes:

- **Acid too strong or too hot** for the grade.
- **Sensitive alloy** (free-machining, martensitic, high-carbon).
- **Chloride contamination** of the bath (poor rinse water or chlorinated cleaners) — a notorious trigger.
- **Insufficient inhibition** — for tough nitric jobs, the right type *adds sodium dichromate* to inhibit attack.



REMEMBER

Flash attack = the acid ate the steel instead of just cleaning it. Darkening, frosting, etching, or rust coming *out* of the bath is a reject, not a bonus. The fix is **gentler conditions for that grade**: lower concentration/temperature, the correct dichromate-inhibited nitric type, or **switch to citric** (far less prone to flash attack). The spec defines which nitric/citric *type/method* is allowed for each alloy — follow it.



HEADS-UP

Brown fumes during flash attack are a real exposure event, not just a quality problem. If a load is flash-attacking, you may be generating **NO_x** — ensure ventilation, protect yourself, and stop to fix the cause. Don't keep re-running the same too-aggressive bath and scrapping parts.

— HOW WE PROVE AN INVISIBLE PROCESS

THE VERIFICATION TESTS IN DEPTH

NO COATING TO SEE — SO WE PROVE IT FIVE WAYS

Because passivation leaves nothing visible, the **test is the deliverable**. Here are the methods named in the specs.

• FREE-IRON / SURFACE TESTS (FAST, COMMON)

- **Copper sulfate test.** A copper sulfate solution is applied (swab or immersion) for a set time. **Free iron causes copper to deposit out of solution as a visible pinkish/copper-colored spot. Copper deposit = free iron present = FAIL.** Fast and sensitive — but **not valid for all grades** (false positives on free-machining/low-chromium grades) and **not for food-contact** parts. Use only where the spec allows.
- **Potassium ferricyanide-nitric acid test (ferroxyl test).** A prepared reagent is applied; **free iron produces blue spots (Prussian blue).** Very sensitive — even tiny iron specks show up. **Blue = free iron present = FAIL.** Reagent is toxic — handle strictly per procedure.

• CORROSION-PERFORMANCE TESTS (SLOWER, DEFINITIVE)

- **Water immersion test.** Parts undergo alternating immersion in (often distilled) water and air over a set number of cycles/hours; **any rust or staining = fail.**
- **High-humidity test.** Parts sit in a controlled high-humidity cabinet for a set time; **rust spots = fail.**
- **Salt spray (ASTM B117).** Parts sit in a sealed cabinet sprayed with salt fog; you record hours until rust appears. **Rust within the spec'd hours = fail.** The most demanding common test.



TIP

A copper sulfate or ferroxyl test takes minutes and catches free iron before a part ever ships. Many shops run a quick free-iron test as an in-process check and reserve salt spray / humidity for periodic lot qualification per the spec.

TEST	SPEED	SENSITIVITY TO FREE IRON	NOTES
Copper sulfate	Fast	High	Grade-limited; not food-contact
Ferroxyl (K-ferricyanide)	Fast	Very high	Toxic reagent; very sensitive
Water immersion	Slow	Moderate	Good general check
High humidity	Slow	Moderate–high	Cabinet required
Salt spray (B117)	Slowest	Overall corrosion	Definitive; per spec hours



REMEMBER

The **copper sulfate** and **ferroxyl** tests look specifically for **free iron** — a colored spot (copper/pink or blue) means iron is still there and the part failed. The **water immersion, humidity, and salt spray** tests check overall corrosion resistance over time. The spec tells you **which** test is required for your part — run *that* one, not whichever is fastest.



TECHNICAL STUFF

ASTM A967 and ASTM A380 describe these test practices in detail, including reagent formulas, exposure times, and acceptance criteria. AMS 2700 ties specific tests to its methods. Acceptance limits (e.g., salt-spray hours, allowable staining) come from the **customer spec**. Always run the **exact method and read against the exact criteria** your job names — treat the details here as orientation, not as the controlling document.



HEADS-UP

Test reagents are real chemicals: **copper sulfate** and **potassium ferricyanide** are toxic, and the ferroxyl reagent involves acid. Mix, use, and dispose of them strictly per procedure and SDS, in PPE — and never combine ferricyanide with strong acid except as the validated reagent recipe directs.

DON'T CONFUSE THESE TWO

PICKLING VS. PASSIVATION

THE MOST IMPORTANT DISTINCTION IN THIS MANUAL AFTER “IT’S NOT A COATING”

Operators (and customers) constantly confuse these two acid processes. They are **not** the same.

	PICKLING	PASSIVATION
Aggressiveness	Aggressive	Mild
Typical acids	Strong — often nitric + hydrofluoric (HNO₃/HF) mix	Dilute nitric or citric
What it removes	Heavy scale, mill scale, weld heat tint, and a layer of the metal itself	Only free iron and light surface contamination
Removes a metal layer?	Yes (etches into the steel)	No (leaves the steel essentially unchanged)
Removes weld scale/heat tint?	Yes	No
Result	A clean, descaled, etched surface	A clean surface with an enhanced passive layer

★

REMEMBER

Pickling is the aggressive acid that removes scale, weld discoloration, and a layer of metal. Passivation is the mild acid that removes only free iron and enhances the passive film. A part with heavy weld scale or heat tint must be pickled (or mechanically descaled) **FIRST** — passivation will *not* remove it. Passivation is the gentle finishing step; pickling is the heavy lifting. Don't expect one to do the other's job.

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HEADS-UP — PICKLING IS EVEN MORE HAZARDOUS

If your shop also pickles, note that **HF (hydrofluoric acid)** is extraordinarily dangerous — it causes deep burns that can be deceptively painless at first and can be **fatal** even from skin exposure, and it requires **calcium gluconate** first aid. Pickling is a separate, higher-hazard process with its own training. **Never treat a pickling tank like a passivation tank.**

💡

TIP

A common customer misconception: “*Just passivate it to get rid of that weld discoloration.*” Passivation can't — heat tint is a chromium-depleted oxide that needs pickling or mechanical removal. If you passivate over heat tint, the part will still corrode at the weld. Flag it and route it correctly.

THE GOVERNING DOCUMENTS

SPECS : ASTM A967, AMS 2700, QQ-P-35

THE DOCUMENTS THAT DEFINE HOW — AND HOW TO PROVE IT

Passivation is governed by industry and customer specifications that define the **method, parameters, and required verification test** by alloy.

- **ASTM A967 / A967M** — *Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts*. Covers both **nitric** and **citric** methods (defined "Nitric" and "Citric" treatment types) and the **test methods** (water immersion, high humidity, salt spray, copper sulfate, ferroxyl, etc.). The most widely cited commercial passivation spec.
- **AMS 2700** — *Passivation of Corrosion Resistant Steels* (aerospace). Defines **Method 1 = nitric** (with several types by concentration/temperature/dichromate) and **Method 2 = citric**. Common in aerospace/defense work.
- **QQ-P-35** — older **federal** passivation spec, now **largely cancelled/superseded** by ASTM A967 and AMS 2700, but still referenced on legacy drawings.
- **ASTM A380** — companion practice for **cleaning, descaling, and passivation** of stainless (includes test procedures and is often cited alongside A967).



REMEMBER

Your job's **traveler/drawing names the governing spec and type**. That spec dictates the **chemistry (nitric or citric), the parameters, and the required acceptance test** for your alloy. **Run exactly what the spec calls out** — don't substitute chemistry or test methods without written approval. If a drawing still calls out **QQ-P-35**, ask your lead which current spec (usually A967 or AMS 2700) it maps to.



TECHNICAL STUFF

These specs are *verify-against* documents: concentrations, temperatures, times, dichromate use, and acceptance criteria vary by **type/method and alloy family**. Treat every parameter in this manual as *representative and educational*; the controlling numbers live in the current revision of the named spec and your supplier's TDS. Always work from the latest revision.



FUN FACT

When AMS 2700 lists "Method 1" and "Method 2," it isn't ranking them — Method 1 is simply nitric and Method 2 is citric. A part passivated by the citric "Method 2" that passes the required test is every bit as airworthy as a nitric one. The spec cares about the *result*, not the acid.

THE QUIET CULPRITS

WATER, RINSE & WASTE

WHERE HIDDEN PROBLEMS — AND REAL LEGAL OBLIGATIONS — LIVE

• WATER & RINSE QUALITY

- **Incoming water hardness** leaves spots and can interfere with rinsing and testing — many lines use **softened or DI water**, especially for the **final rinse** before drying.
- **Chlorides in rinse water** are a flash-attack and pitting risk — another reason to control rinse-water quality.
- **Monitor rinse conductivity** as your early-warning gauge for acid/cleaner drag-in.

**TIP**

If you're chasing spots, streaks, or inconsistent test results and the chemistry checks out, **suspect the rinse water**. Hard or chloride-laden rinse water is a quiet, common culprit.

• WASTE & THE ENVIRONMENT

Passivation has real environmental obligations — and here the **nitric vs. citric** choice matters a lot:

- **Nitric waste:** spent baths and rinses carry **nitrate** (a regulated nutrient that feeds algae blooms) and **acidity** — requiring neutralization and permitted discharge. If **dichromate** is used, the waste also contains **hexavalent chromium**, a hazardous waste requiring **chromium reduction/treatment**.
- **Citric waste:** **biodegradable** and far easier to treat — a major environmental advantage and a key driver of the shift to citric.
- **Spent alkaline cleaners and skimmed oils** are waste streams too.

**HEADS-UP**

Never dump any bath, rinse, or skimmed oil down a drain on your own judgment. Discharges are permitted and treated. **Follow your shop's waste procedures and your wastewater permit** — violations carry serious legal and environmental consequences. **Nitric/nitrate, chromium, and acid discharges are all regulated.**

**FUN FACT**

Part of why citric passivation took off isn't just worker safety — it's the drain. Swapping nitric for citric can turn a regulated, nitrate- (and sometimes chromium-) bearing waste stream into a far more benign, biodegradable one, simplifying permits and treatment.

— HOW WE PROVE IT WORKED

QUALITY CHECKS

PASSIVATION IS JUDGED BY TESTS, NOT BY APPEARANCE

CHECK	WHAT IT TELLS YOU	HOW
Water-break test	Is the part truly clean <i>before</i> the acid?	Watch water sheet vs. bead (free, every load)
Visual (limited)	Is there flash attack/staining?	Darkening/etching = reject; otherwise looks unchanged
Copper sulfate / ferroxyl	Free iron present?	Colored spot (copper or blue) = fail
Water immersion / high humidity	Corrosion resistance over time	Rust/staining = fail
Salt spray (ASTM B117)	Overall corrosion resistance	Rust within spec'd hours = fail



REMEMBER

Passivation is judged by **what the surface does in a corrosion test**, not by what it looks like. A clean appearance is necessary but **not sufficient** — only the **spec-required test** confirms the part is free of free iron and properly passive. The water-break test up front and the verification test at the back are the two checks that bracket every job.



TECHNICAL STUFF — WHAT IS “SALT SPRAY”?

It's an accelerated corrosion test (ASTM B117): samples sit in a sealed cabinet sprayed with salt fog, and you record how many **hours** until rust appears. More hours = better corrosion resistance. Customers use it to specify and verify corrosion requirements on the finished part.



TIP

Run a **control/witness sample** of the same grade through the line with significant loads and test it. It's a cheap insurance policy: if the control fails, you catch a process drift before a whole lot of customer parts ships un-passivated.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

SYMPTOM	LIKELY CAUSE(S)	FIRST THINGS TO CHECK
Free iron remains (fails copper-sulfate/ferroxyl)	Inadequate clean; bath too weak/cold/short; carbon-steel cross-contamination	Re-check cleaning & water-break; verify bath conc/temp/time; check fixtures (no carbon steel)
Rust after passivation / before shipping	Free iron not removed; acid dried on part; poor rinse/dry	Re-clean & re-passivate; rinse promptly; DI final rinse + dry
Flash attack (darkened/etched/frosted)	Acid too strong/hot for grade; sensitive alloy; chloride contamination	Gentler type (lower temp / dichromate / citric); check chlorides; confirm grade
Brown fumes off the bath	Over-aggressive nitric reaction / flash attack	Ventilate; investigate bath strength/temp & grade; consider citric
Spots / streaks / staining	Hard or chloride rinse water; acid dried; air-dried	Soften/DI rinse water; rinse & dry promptly
Failed salt spray / humidity	Free iron remaining; wrong chemistry/type for grade; poor cleaning	Re-clean; verify spec type for the alloy; re-test control sample
Weld area still corrodes	Heat tint/scale not removed — needs pickling, not passivation	Route to pickling/descale first, then passivate
Citric bath cloudy/smelly/weak	Microbial growth / depleted bath	Treat or refresh per TDS; maintain bath control



TIP

The fastest diagnostic instinct: **was it clean, and is it the right process for that surface?** Most failures are either (1) inadequate degreasing (the part wasn't clean enough to passivate) or (2) someone asked passivation to do **pickling's** job (remove scale/heat tint). After those, look at bath strength/temperature and the grade.



HEADS-UP

Before you change bath concentration or temperature, confirm the change with your lead and the spec/TDS — and remember **nitric is an oxidizer** and additions can drive **flash attack and NOx**. Bath adjustments are made only by authorized personnel, in full PPE, with ventilation running.

— YOUR DECODER RING

GLOSSARY — EVERY TERM, PLAIN AND SIMPLE

ANYTIME A WORD IN THIS MANUAL MADE YOU PAUSE, FLIP BACK HERE

AMS 2700: Aerospace passivation spec; Method 1 = nitric, Method 2 = citric.

ASTM A967: Main commercial spec for chemical passivation; covers nitric and citric methods and the test methods.

ASTM A380: Companion practice for cleaning, descaling, and passivation of stainless.

ASTM B117: Salt-spray (salt-fog) corrosion test.

Austenitic stainless (300-series): Common, forgiving, high-chromium/low-carbon stainless (e.g., 304, 316); passivates easily.

Chelant / chelator: A chemical (like citric acid) that grabs and holds metal ions (iron) in solution so they can't redeposit.

Chromium oxide: The chromium-rich oxide that forms the passive layer; the source of stainless's corrosion resistance.

Citric acid passivation: Modern, biodegradable, safer chemistry; no NOx, no Cr(VI); increasingly the standard.

Conversion coating: A process that turns the surface into a *new* compound (e.g., iron phosphate). **Passivation is NOT this** — it adds nothing.

Copper sulfate test: Free-iron test; copper-colored deposit = free iron = fail. Grade-limited; not food-contact.

Cr(VI) / hexavalent chromium: Carcinogenic chromium found in sodium dichromate (added to some nitric baths); OSHA 29 CFR 1910.1026.

DI water (deionized): Water with minerals removed; rinses cleaner, fewer spots.

Drag-out / drag-in: Solution clinging to a part leaving a tank (drag-out) and arriving at the next (drag-in) — what rinses fight.

Ferroxyl test (potassium ferricyanide-nitric acid): Very sensitive free-iron test; blue spots = free iron = fail. Toxic reagent.

Flash attack (flash etching): Over-aggressive acid attack on the steel itself; surface darkens/etches/rusts — a defect.

Free-machining stainless (303, 416): Sulfur-added grades that machine easily but are flash-attack-prone.

Free iron: Ordinary (non-stainless) iron contamination left on the surface from tooling/handling; it rusts and triggers pitting — the thing passivation removes.

Heat tint: Dark, chromium-depleted oxide near welds; **removed by pickling, NOT passivation.**

LOTO (Lockout/Tagout): De-energizing and locking out equipment before maintenance.

Martensitic stainless (410, 420, 440): Higher-carbon, hardenable grades; flash-attack-prone; need gentler conditions.

Nitric acid passivation: Classic chemistry; strong oxidizer; can evolve toxic NOx fumes; sometimes uses sodium dichromate.

NOx (nitrogen oxides): Toxic brown/red-brown fumes from nitric acid, esp. during flash attack; serious inhalation hazard with delayed lung effects.

Passivation: Chemical treatment that removes free iron and enhances the stainless's own chromium-oxide passive layer — **no coating added.**

Passive layer / passive film: The thin, self-healing chromium-oxide skin that makes stainless "stainless."

Pickling: Aggressive acid (often HNO₃/HF) that removes scale, heat tint, and a layer of metal — the heavy-duty cousin of passivation.

QQ-P-35: Older federal passivation spec, largely superseded by ASTM A967 / AMS 2700.

Salt spray: See ASTM B117.

Sodium dichromate: Cr(VI) additive used in some nitric baths to inhibit flash attack on tough grades; a carcinogen.

Stainless steel: Iron alloy with ≥ ~10.5% chromium, which forms the protective passive layer.

Substrate: The base metal (here, stainless steel) being treated.

TDS / SDS: Technical Data Sheet (how to run a product) / Safety Data Sheet (its hazards). Always follow both.

Water-break test: Free cleanliness check: water sheets off a clean part, beads on an oily one.

— TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO



REMEMBER

An operator is “line-qualified” only when every station they run is signed. Nobody works the **nitric bath**, handles a **dichromate (Cr(VI)) bath**, or handles **toxic test reagents** solo until the **Safety / PPE & SDS** row is signed. Safety qualification comes first, always.

Sign-off levels: **O** = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	Safety / PPE & SDS (all stations)	_____	_____	_____	_____
2	Nitric acid & NOx hazard awareness	_____	_____	_____	_____
3	Cr(VI) program (if dichromate in use)	_____	_____	_____	_____
4	Station 01 — Inspect & Load (grade ID, no carbon-steel fixtures, scale = STOP)	_____	_____	_____	_____
5	Station 02 — Clean / Degrease + water-break test	_____	_____	_____	_____
6	Station 03 — Rinse (post-clean) + conductivity	_____	_____	_____	_____
7	Station 04 — Passivate (nitric/citric); flash-attack recognition	_____	_____	_____	_____
8	Bath control — concentration/temperature/time	_____	_____	_____	_____
9	Station 05 — Rinse (post-passivate, DI final)	_____	_____	_____	_____
10	Station 06 — Dry (no spots, clean-glove handling)	_____	_____	_____	_____
11	Station 07 — Test & Verify (copper sulfate / ferroxyl / salt spray / humidity)	_____	_____	_____	_____
12	Pickling vs. passivation — knowing the difference	_____	_____	_____	_____
13	LOTO (heaters/pumps/hoists/dryer)	_____	_____	_____	_____
14	Spec literacy (ASTM A967 / AMS 2700 type by grade)	_____	_____	_____	_____
15	Emergency response — eyewash, shower, fume exposure, spill	_____	_____	_____	_____
16	Waste / drag-out / nitrate & chromium handling	_____	_____	_____	_____
17	Troubleshooting — symptom recognition	_____	_____	_____	_____

Operator name: _____ Hire/start date: _____
Supervisor name: _____ Review cycle: ☐ Annual ☐ Other: _____



HEADS-UP

Safety rows (1, 2, 3, 13, 15) are not optional and not “fill in later.” An operator may not work *any* station until PPE/SDS, **nitric/NOx awareness**, **Cr(VI) program (if applicable)**, LOTO, and emergency response are signed. Safety competency comes before production competency, always.

PART FOUR — BACK MATTER & CERTIFICATION

20 QUESTIONS • PASSING SCORE 80% (16/20)

PASSIVATION COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

**REMEMBER**

Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (Q5, Q6, Q15, Q19) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: _____ Date: _____ Score: _____ / 20 **Safety-gate Qs (all correct): 5, 6, 15, 19**

Q1. Passivation works by:

- A. Electroplating a thin chromium layer onto the part B. Baking a phosphate powder onto the stainless C. Chemically removing free iron and letting the stainless's own chromium-oxide passive layer reform stronger — no coating is added D. Spraying a protective polymer film on the surface

Q2. (Short answer) In one or two sentences, explain why passivation is **not a coating**. What actually happens at the surface?

Q3. Stainless steel's corrosion resistance comes from:

- A. A wax sealer applied at the mill B. A self-healing chromium-rich oxide passive layer the steel forms on its own C. The nickel content only D. The passivation acid leaving a film behind

Q4. (Short answer) What is **free iron**, where does it come from, and why does it ruin otherwise-good stainless?

Q5. [SAFETY GATE] Nitric acid passivation can evolve which hazardous fumes, especially during flash attack?

- A. Brown/red-brown NOx (nitrogen oxides) — a serious inhalation hazard with delayed lung effects B. Harmless water vapor only C. Chlorine gas D. Carbon dioxide

Q6. [SAFETY GATE] You see **brown/red-brown fumes** rising off the nitric tank while a load is processing. This most likely means ____ and you should ____:

- A. The bath is fresh; add more parts B. The lights are too bright; dim them C. It's normal steam; ignore it D. The acid is attacking the steel (flash attack) and generating NOx — ensure ventilation, don't breathe it, and report it

Q7. (Short answer) Why is **degreasing/cleaning (Station 02)** described as the make-or-break step? What happens if you passivate an oily part?

Q8. **Free iron** on stainless typically comes from:

- A. The chromium in the alloy B. Carbon-steel tooling, steel wool, machining/forming, or contact with carbon steel C. The passivation acid itself D. Deionized water

Q9. (Short answer) Give **two** advantages of **citric** passivation over **nitric** passivation.

Q10. **Pickling** differs from passivation in that pickling:

- A. Adds a chromium coating B. Is gentler than passivation C. Is aggressive — it removes heavy scale, heat tint, and a layer of the metal itself D. Is only used on aluminum

Q11. (Short answer) What is **flash attack**, and name two conditions or grades that make it more likely.

Q12. Which stainless grades are **most prone to flash attack** and need gentler conditions?

A. Free-machining (e.g., 303, 416) and high-carbon/martensitic (e.g., 410, 420, 440) B. Only austenitic 304 and 316 C. All grades equally D. No grades are ever sensitive

Q13. The main industry specifications for chemical passivation of stainless are:

A. ASTM B117 and D3359 only B. There are no specs for passivation C. ASTM A967 and AMS 2700 (older: QQ-P-35) D. OSHA 1910.1026 only

Q14. (Short answer) A part comes in with **heavy weld scale and dark heat tint**. Should you passivate it as-is? Why or why not — and what should happen first?

Q15. [SAFETY GATE] Sodium dichromate, sometimes added to nitric baths for tough grades, is:

A. A harmless thickener B. Hexavalent chromium (Cr(VI)) — a confirmed human carcinogen with a special OSHA program (29 CFR 1910.1026) C. A type of citric acid D. A rinse aid

Q16. Because passivation leaves **no visible coating**, how do you confirm a part was successfully passivated?

A. It will turn gold B. It will feel slippery C. By weighing it D. By running the spec-required verification test (copper sulfate, ferroxyl, salt spray, humidity, or water immersion)

Q17. (Short answer) In plain terms, explain how the passivation acid does **two jobs at once** — removing free iron *and* enhancing the chromium-oxide passive layer.

Q18. A stainless part comes out of the bath **darkened, frosted, and etched** instead of bright and clean. This is:

A. Flash attack — a defect; the acid was too aggressive for the grade (use gentler conditions or citric) B. A sign of extra-good passivation C. Normal and desirable D. Proof a coating was deposited



SAFETY SECTION — MUST BE CORRECT TO CERTIFY

Questions 5, 6, 15, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

Q19. (Short answer / Safety) [SAFETY GATE] List **four** pieces of PPE you must wear at the **nitric passivation** station, name the **one** **ventilation/fume control** required because of **NOx**, and name **one** class of materials nitric acid must be kept **away from**.

Q20. (Short answer) Name **two** honest **limitations** of passivation (things it cannot do).

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD



HEADS-UP

Keep this page out of the trainee's copy. Questions **5, 6, 15, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Self-check answer distribution (the 11 multiple-choice items): A = 3, B = 3, C = 3, D = 2. If your key clusters on one letter, you've mis-keyed.

Q1. C — Removes free iron and lets the stainless's own chromium-oxide passive layer reform; **no coating is added**.

Q2. Nothing is deposited or created as a new material. The acid **dissolves free iron/contamination** off the surface, leaving it **chromium-rich**, and the steel's **own chromium-oxide passive film reforms thicker/more uniform**. The protection comes from the stainless itself.

Q3. B — The self-healing **chromium-rich oxide passive layer** the steel forms on its own.

Q4. Free iron is **ordinary (non-stainless) iron contamination** left on the surface from **carbon-steel tooling, steel wool, machining/forming, or contact with carbon steel**. It has no chromium, so it **rusts**, and it can **trigger pitting/localized corrosion** in the good stainless around it.

Q5. A — Brown/red-brown **NOx (nitrogen oxides)**; serious inhalation hazard with **delayed** lung effects.

Q6. D — The acid is attacking the steel (**flash attack**) and generating **NOx**; ensure ventilation, don't breathe it, report it.

Q7. The passivating acid must touch **bare stainless** to dissolve free iron; **oil blocks it completely**. Passivate an oily part and you've **passivated nothing** — it may look done but still rusts/fails the test.

Q8. B — Carbon-steel tooling, steel wool, machining/forming, or contact with carbon steel.

Q9. Any two of: **biodegradable / greener; much safer / less corrosive; no NOx fumes; no Cr(VI); less flash-attack-prone; easier waste handling**.

Q10. C — Pickling is **aggressive** — it removes heavy scale, heat tint, and a **layer of the metal itself** (passivation is mild and removes only free iron).

Q11. Flash attack = the acid **aggressively attacks the stainless itself** (not just free iron), leaving the surface **darkened/frosted/etched/rusty** (a defect). Likely causes (any two): **acid too strong/hot, free-machining grades (303/416), martensitic/high-carbon (410/420/440), chloride contamination, insufficient inhibition**.

Q12. A — Free-machining (303, 416) and high-carbon/martensitic (410, 420, 440).

Q13. C — ASTM A967 and AMS 2700 (older: QQ-P-35).

Q14. No — **do not passivate it as-is**. Passivation will **not** remove heavy scale or heat tint. It must be **pickled (or mechanically descaled) first**, then passivated.

Q15. B — Hexavalent chromium (Cr(VI)) — a confirmed carcinogen; OSHA 29 CFR 1910.1026 program.

Q16. D — By running the **spec-required verification test** (copper sulfate, ferroxy/potassium ferricyanide, salt spray, humidity, or water immersion). There's no coating to see.

Q17. The acid **dissolves free iron and contamination** off the surface (job 1), which leaves it **chromium-rich**; the acid/oxidizer (or, for citric, chelation plus oxygen) then helps the **chromium-oxide passive film reform thicker and more uniform** (job 2). No coating is added — it's the steel's own film, optimized.

Q18. A — **Flash attack**, a defect; the acid was too aggressive for the grade. Use gentler conditions (lower temp/concentration, dichromate-inhibited type) or **citric**.

Q19. PPE (any four): sealed chemical splash goggles, face shield, **nitric-rated** chemical-resistant gloves, chemical apron, chemical-resistant/non-slip boots, respirator if ventilation is inadequate. **Fume control: local exhaust ventilation** (to capture NOx/acid mist). **Keep nitric away from: organics / solvents / rags / reducing agents / combustibles** (violent reaction/fire).

Q20. Any two of: **it's not a coating and adds no thickness/wear layer; it can't remove weld scale or heat tint (that's pickling); it can't fix a bad/under-alloyed or sensitized stainless; it can't help non-stainless metals; results aren't visible — must be tested; nitric route has fume/waste hazards**.



REMEMBER

16/20 passes — but every safety question (5, 6, 15, 19) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. We don't gamble with people — and **nitric and its NOx fumes are exactly why**.

— PRINT, FILL IN, SIGN



PLATING POSTERS

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CERTIFICATE OF
COMPLETION
PASSIVATION (STAINLESS STEEL) TRAINING PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Passivation (Stainless Steel)** training course — covering all seven process stations (Inspect/Load, Clean/Degrease, Rinse, Passivate, Rinse, Dry, and Test/Verify), the fundamentals of passivation as a **chemical cleaning and enhancement process that adds no coating**, the role of **free iron** and the **chromium-oxide passive layer**, **nitric vs. citric** chemistries, **grade-dependence and flash attack**, the **pickling-vs-passivation** distinction, the **verification tests** (copper sulfate, ferroxyl, salt spray, humidity, water immersion), the governing specs (**ASTM A967, AMS 2700, QQ-P-35**), troubleshooting, and the safety practices for **nitric acid and its NOx fumes**, hexavalent chromium (where dichromate is used), hot acid baths, degreasers, and toxic test reagents — and has passed the Completion Test with a score of _____ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: _____ Re-certification due: _____

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer and industry specifications (ASTM A967 / AMS 2700), and applicable regulatory requirements before production use. Nitric acid is a strong oxidizer that can evolve toxic NOx fumes; hexavalent chromium, where used, is a confirmed human carcinogen — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.

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